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MILITARY SPECIFICATION

OVENS, BAKING AND ROASTING, FORCED CONVECTION, GAS

This limited coordination military specification has been prepared by the US Army Natick Research and Development Laboratories, based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-O-43679. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers one and two compartment gas fired, forced convection baking and roasting ovens.

* 1.2 Classification. Ovens shall be of the following types, style, and model as specified (see 6.2).

Type I - One compartment
Type II - Two compartment
Style 1 - Stationary
Model B - With removable shelf supports and removable shelves

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-698 - Steel, Sheet and Strip, Low-Carbon

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7310

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MILITARY

MIL-V-173 - Varnish, Moisture, and Fungus-Resistant (For the Treatment of Communications, Electronic and Associated Electrical Equipment)
MIL-B-2316 - Baking Equipment, Including Units, Assemblies, Repair Parts, and Tools, Preparation for Delivery Of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement Of

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN GAS ASSOCIATION, (A.G.A.)

Directory of Certified Appliances and Accessories

(Application for copies should be addressed to the American Gas Association, 8501 East Pleasant Valley Road, Cleveland, OH 44131.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

Z 21.28 - Commercial Gas Baking and Roasting Ovens
Z 21.41 - Quick Disconnect Device for Use with Gas Fuel
Z 21.45 - Flexible Connectors of Other than All-Metal Construction for Gas Appliances

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip
A 176 - Stainless and Heat-Resisting Chromium Steel Plate Sheet, and Strip
A 412 - Stainless and Heat-Resisting Chromium-Nickel-Manganese Steel Plate, Sheet, and Strip
A 463 - Steel Sheet, Cold-Rolled, Aluminum-Coated Type 1

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL SANITATION FOUNDATION (NSF)

Standard No. 4 - Commercial Cooking and Hot Food Storage Equipment

Listing of Food Service Equipment

(Application for copies should be addressed to the National Sanitation Foundation, NSF Building, Ann Arbor, MI 48105.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. The oven shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product with any added features needed to comply with the requirements of this specification. Modifications to add features shall not incorporate different parts unless such parts are used on other current commercial models. Standard or modified commercial products furnished in accordance with this specification shall be identifiable by all regular manufacturer's or commercial service organizations servicing the brand involved. Service organizations shall be capable of providing complete parts and repair services on models furnished to the government consistent with their normal commercial practices.

* 3.2 Standards. The convection ovens shall conform to the applicable requirements of NSF Standard No. 4 and ANSI Standards Z 21.28, Z 21.41 and Z 21.45.

3.2.1 Compliance. Prior to commencing production the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative that the convection ovens he proposes to furnish under this specification meet the applicable requirements of NSF Standard No. 4 and ANSI Standards Z 21.28, Z 21.41 and Z 21.45 as follows:

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* 3.2.1.1 NSF. Acceptable evidence of meeting the requirements of NSF shall be one of the following:

(1) A listing in the current edition of the NSF "Listing of Food Service Equipment" and display of the NSF seal on the finished convection oven.

(2) A certification for the convection ovens issued by NSF under their special one-time contract evaluation/certification service.

(3) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the convection ovens are being procured, indicating that the convection ovens have been tested and conform to NSF Standard No. 4.

* 3.2.2 ANSI certification. Acceptable evidence of meeting the applicable requirements of Z 21.28, Z 21.41 and Z 21.45 shall be a reproduced copy of the American Gas Association (A.G.A.) Appliance Certificate, a listing in the latest A.G.A. "Directory of Certified Appliances and Accessories", or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating compliance with ANSI Standards Z 21.28, Z 21.41 and Z 21.45.

* 3.3 Materials. Materials (see 6.4), and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer for his standard commercial convection oven, provided the completed item complies with all provisions of this specification.

3.3.1 Steel.

3.3.1.1 Stainless and heat-resisting steel. The stainless and heat-resisting steel shall conform to type 202 of ASTM A 412; type 302 or 304 of ASTM A 167; or any of 400 types of ASTM A 176.

3.3.1.2 Carbon steel. Carbon steel shall be commercial quality (CRCQ) conforming to QQ-S-698.

3.3.1.3 Aluminum-coated steel. The aluminum-coated steel shall conform to type I of ASTM A 463.

* 3.4 Design. Convection ovens shall be designed for baking and roasting and shall incorporate a two speed oven blower to provide for circulation of forced air throughout the oven compartment. Type I and II ovens shall have a single oven compartment and two oven compartments, respectively. The type I oven shall consist essentially of a body, oven chamber linings, door(s) with handle(s), shelves, shelf support (model B), heating units, two speed oven blower, temperature control, timer, oven light, burner on-off control, blower speed selector switch (low and high), heat tempered glass viewing door(s) and flue riser. The type II oven shall consist of two type I ovens for stacking and securing above one another. Each compartment shall be capable of being operated independently and shall be individually controlled by a thermostat. The convection oven hourly gas input rating shall not exceed 115,000 British thermal units (Btu) for type I ovens and 230,000 Btu for type II ovens.

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3.4.1 Style 1, stationary. The type I, style 1 oven shall be mounted on legs directly attached to the four bottom corners of the oven or mounted on an open or enclosed stand. The type II, style 1 oven shall be mounted on legs directly attached to the four bottom corners of the oven. All style 1 ovens shall be provided with leveling devices.

3.4.2 Dimensions. Convection ovens shall be within the dimensional limits shown in table I.

TABLE I. Dimensions (inches)

Measurements	Minimum	Maximum
Width overall	33	52
Depth overall	30	48
Height 1/ overall (excluding vent)		
Type I	26	63
Type II	52	78
Height 1/ to bottom deck		
Type I	26	38
Type II	11	19-3/4
Height 1/ to top deck (type II)	30	51-3/4
Compartment (inside usable)		
Width	21-1/2	29-1/4
Depth	20	30
Height	19-3/8	28

1/ Height from floor, including supporting legs and base.

NOTE: When the blower motor is not an integral part of the oven, the motor encasement shall not extend beyond 9 inches on the side opposite the oven controls.

* 3.4.3 Model B. Model B ovens shall be provided with removable shelf supports and removable shelves.

3.5 Construction. Type I and II oven components fabricated of stainless steel, carbon steel, or aluminized steel shall conform to 3.3.1.1, 3.3.1.2, or 3.3.1.3, respectively.

3.5.1 Body. The body shall be of a frame or shell construction. When frame construction is used, the frame shall consist of a base frame, top frame, and vertical angles to form a rigid assembly. Exterior sheets shall be welded to the frame, inclosing the frame construction therein. When of shell construction, the outer shell shall be stiffened and reinforced. The front of the ovens shall be provided with means for mounting the controls.

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3.5.1.1 Type I. The exterior surfaces of the type I oven except the bottom and back shall be fabricated of stainless steel. The bottom and back surface may be fabricated of carbon steel or aluminized steel.

3.5.1.2 Type II. The exterior surfaces of the type II oven (front, sides, and top) shall be fabricated of stainless steel. The bottom, top, and back of the lower oven, and the bottom and back of the upper oven shall be fabricated from carbon steel or aluminized steel. The top surface of the upper oven may be fabricated of carbon steel provided the exterior surface is stainless steel.

* 3.5.2 Oven compartment linings. Each oven compartment shall be provided with oven chamber linings fabricated from stainless steel. When specified (see 6.2), the oven baking compartment shall be fabricated from aluminized coated steel specified in 3.3.1.3, or porcelain over carbon steel (see 3.3.1.2). The oven chamber linings shall be provided without open joints and shall be reinforced. Removable oven liner panels fabricated from stainless steel shall be provided. The removable oven linings shall consist of top, right-hand, left-hand and bottom panels.

* 3.5.2.1 Vent or air baffle. Vent or air baffles, when provided, may be fabricated from stainless steel specified in 3.3.1.1. When specified (see 6.2), vent or air baffles, when provided, shall be fabricated of aluminum-coated steel specified in 3.3.1.3 or porcelain enamel over carbon steel (see 3.3.1.2).

* 3.5.3 Doors. The door(s) shall be of the drop, counter-balanced type, twin counter-balanced type, or vertically split, side opening type. Drop door(s) shall be balanced so that when in the fully open position it will remain fully open without securing, permitting full access to the usable oven area. Vertically split doors when in the fully open position shall open flush with the oven interior sidewalls and shall not obstruct the removal of the shelves, and when applicable, shelf supports, shelf carrier, or shelf carrier dolly. Handle(s) shall be provided on the door(s). When one handle is provided on a convection oven with two doors, both doors shall operate simultaneously when opened and may close simultaneously. Heat-resisting viewing window(s) shall be located in the door(s) so that at least the center three shelves may be viewed when the door(s) are closed. When specified (see 6.2), doors shall be without windows. Each convection oven shall be provided with a latch, clip, or other means to hold the door(s) in the closed position. The oven door interior shall be fabricated from stainless steel. When specified (see 6.2), the oven door interior shall be fabricated from aluminized steel, or shall be porcelain enamel on carbon steel. The convection oven may be provided with a door control switch.

* 3.5.4 Shelves. Each oven compartment shall be provided with a minimum of 5 equally spaced removable shelves supported by removable shelf supports for model B ovens. Shelves shall be fabricated of steel rod, steel frame with steel slats, or expanded steel. Shelves shall not jam or buckle when placed into, or removed from, the shelf supports. Each shelf shall be capable of supporting a load of 20 pounds per square foot without more than 1/8 inch deflection, dislodgement, or permanent deformation when tested as specified in 4.4.1.

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3.5.5 Venting. A means shall be furnished for venting compartments. When a flue riser is provided, it shall be located at the rear of the oven and a single flue shall vent both the upper and lower oven compartments on type II ovens.

* 3.5.6 Fuel system. The fuel system, including burners, shall be designed to efficiently burn natural or liquified petroleum gas (LPG) with appropriate conversion.

* 3.5.6.1 Natural gas. The unit shall be furnished to operate on natural gas. Main burners shall be provided with fixed orifices with orifice hoods or spuds that can be interchanged. The orifice hoods or spuds installed for natural gas shall be unpainted brass. An automatic pilot including a 100 percent shut-off shall be provided on each unit.

* 3.5.6.2 LPG. The contractor shall furnish black color-coded orifice hoods or spuds to convert the unit to operate on LPG. The contractor shall also furnish complete instructions for conversions to operate on LPG.

* 3.5.6.3 Automatic ignition. All main burners shall be equipped with means for automatic ignition of the gas. A pilot, including a 100 percent shut-off device shall be provided for each oven burner. The pilot shall automatically shut off the gas to all oven burners including oven pilot burners, in the event that means of ignition becomes inoperative.

* 3.5.6.4 Burner unit(s). Each oven compartment shall be provided with independently operated burner unit(s) located to provide even heat distribution throughout the oven compartment. The burner unit(s) shall be fully protected and removable from the front. Heat distribution in the baking chamber shall not vary more than specified in 4.4.5, when tested in accordance with 4.4.5.

* 3.5.6.5 Accessories. A pressure regulator for natural gas shall be furnished with each unit. The unit shall be furnished with a connector conforming to Z 21.45 and a quick disconnect conforming to Z 21.41. The connector length shall be 60 inches.

3.5.7 Electrical characteristics. Unless otherwise specified (see 6.2), the electrical controls, lights, switches and blower shall be suitable for operation on a nominal 120-volt, 60-hertz, 1-phase (ac) circuit. All controls, switch(es), and indicator light(s) shall be located in a control panel at the front of the oven.

* 3.5.7.1 Oven blower. Each oven compartment shall have a two speed blower. The oven blower shall provide even circulation and heat distribution throughout the oven compartment when tested as specified in 4.4.5. The oven blower shall operate whenever the door(s) are closed. A separate oven blower control switch may be provided for cooling. The oven blower motor shall be of the capacitor or split phase type. The oven blower shall be adequate to sustain continuous operation at high and low oven blower motor speed settings. Motor speed on the high speed setting shall be nominal 1725 rpm and the low speed setting shall be between 750 and 1125 rpm. Switching from one speed to the other shall be accomplished without damage to the motor and to the motor electrical system. A protective device shall be provided over the blower to restrict roasting pan handle, when in the raised position, from hitting the blower. Oven blower wheel shall be fabricated of stainless steel as specified in 3.3.1.1 or other material suitable for the purpose intended. If stainless steel is not used the blower wheel shall be permanently coated with a material to prohibit corrosion.

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3.5.7.2 Oven light. The oven shall be provided with light(s) to illuminate the oven interior. The light(s) shall be recessed or protected with a heated tempered glass window(s). The oven light(s) may be actuated by a manual switch or automatically by a door switch. The light and protective window shall be accessible for light replacement and window cleaning.

3.5.7.3 Indicator light. Ovens shall be furnished with a separate indicator light that glows when the oven is operating and goes out whenever the proper temperature has been attained.

3.5.7.4 Operation. Ovens shall not leak gas, and burners, controls and moving parts shall operate and perform as intended when tested as specified in 4.4.2.

3.5.8 Temperature control. Each oven compartment shall have a manually operated thermostat to provide automatic heat control of the oven compartment and permit cooking operations to be accomplished by maintained temperature. The thermostat shall be capable of maintaining any oven temperature within the range of 250°F to at least 400°F, plus or minus 13°F of the temperature setting when tested as specified in 4.4.3. The thermostat actuating components shall be protected against moisture, grease, and direct heat radiation. An "OFF" position shall be marked on the thermostat dial or a separate switch shall be provided. Temperature control dial shall be marked from 150°F to at least 500°F.

3.5.8.1 Preheat time. The oven shall attain a compartment temperature of 400°F within 20 minutes in an ambient temperature of 70°F plus or minus 5°F, when tested as specified in 4.4.4.

3.5.9 Oven timer. Each oven compartment shall be provided with a 60 minute oven timer installed in the control panel at the oven front. The oven timer shall be equipped with an alarm signal which will sound when the preset time has been completed. The timer range shall be 0 to 60 minutes.

3.5.10 Terminal board or box for style 1 convection ovens. A terminal board or box shall be furnished on style 1 ovens. The board or box shall be within the oven enclosure and shall provide for permanent connection to the installation wiring system. The terminal board or box shall have a grounding terminal with connection to the oven outer body.

* 3.5.11 Electromagnetic compatibility. When specified (see 6.2), ovens shall be designed and equipped for electromagnetic compatibility in accordance with part 10, class C3, group I of MIL-STD-461 (see 4.4.6).

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3.5.12 Fungus and moisture resistance. When specified (see 6.2), the oven shall be treated to resist fungus and moisture as follows:

(a) Circuit elements with temperature rise not more than 75°F: Circuit elements, which have a temperature rise of not more than 75°F when operating at full load, shall be coated with fungus-resistant varnish conforming to MIL-V-173. Circuit elements include, but are not limited to, cable, wire, terminal blocks, capacitors, and coils.

(b) Electrical components not requiring fungus moisture resistance: Electrical components such as switches, fuses, and contacts, shall not be treated. Other materials and components which are inherently fungus resistant or are protected by hermetic sealing need not be treated.

3.6 Mounting.

3.6.1 Stand and legs. All legs shall be fabricated of stainless steel. All legs except on the open stand shall be removable. Open stands and the front, bottom and side panels of the enclosed stands shall be fabricated of stainless steel. Open and enclosed stands shall be reinforced and their extremities shall not protrude beyond the extremities of the oven exterior.

3.7 Marking.

* 3.7.1 Identification plate. The convection ovens shall be permanently marked with the serial number, model number and the manufacturer's name, trade name, or trademark of such known characters to readily identify the manufacturer.

* 3.7.2 Users' instruction plate. Each type and style of oven shall be equipped with a users' instruction plate, permanently fastened to the outside front of the unit. The plate shall be constructed of corrosion-resistant metal having a minimum thickness of 0.0239 inch, and shall have the following information etched or stamped thereon in legible characters. The word "CONVECTION OVEN" shall be in characters not less than 3/16 inch high.

CONVECTION OVEN

USERS' INSTRUCTIONS

1. Turn gas burner controls on and adjust thermostat to desired oven temperature.

2. Oven. If oven burner fails to light, due to failure of pilot flame, manually shut gas off and wait 5 minutes with combustion compartment door open, then reignite the automatic pilot according to manufacturer's instructions. Use thermostat to control oven heat. Turn gently, do not force.

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3. Valves. Keep all burner valves clean and well lubricated. Lubricate burner valves with valve lubricant furnished with oven at least once every three months. For additional supply of valve lubricant, requistion Grease, Plug, Valve, Silicon Base, FSN 9150-935-3992.

4. Load shelves evenly, making sure food does not contact oven linings.

5. Do not overload oven compartment.

6. Clean oven daily or if necessary more often. Clean oven exterior and interior with damp cloth or mild detergent and wipe dry. Scrape all carbonized matter from door, door frame, and outside lower lip of door using a scraper. Clean shelves daily or if necessary more often.

3.8 Repair and maintenance. Provisions shall be made to allow ready adjustment, service, and replacement of gas system components and operational electrical accessories and components, without disturbing or disassembly of any other unrelated major component or equipment.

3.9 Finish. Ovens shall be smooth on all surfaces and edges and shall be free from discoloration or stains. Oven compartments shall be finished in accordance with the manufacturer's standard practice for stainless steel, aluminized steel, or porcelain enamel on carbon steel, as applicable.

3.10 Workmanship. All components and assemblies of the oven shall be free from dirt and extraneous material, burrs, slivers, rough die, gouge, tool, trim and grind marks, dents and cracks. Castings, and molded parts if used, shall be free of sand, fins, pits, blowholes, and sprues. External surfaces for all components and assemblies shall be free from sharp edges and sharp corners, except when corners are required.

3.10.1 Metal fabrication. Metal used in fabrication of the oven shall be free from kinks. Forming and shearing shall not cause damage to the metal and shall be done neatly and accurately. Corners shall be square and true and all bends of a major nature shall be made with dies of fixtures to insure uniformity of size and shape.

3.10.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease and other foreign matter. Welds shall be smooth and free from cracks, burn holes, undercuts or incompleted fusion. All scale and flux shall be removed from the finish weld area.

3.10.3 Fastening devices. Threaded fasteners and rivet holes shall be accurately punched or drilled and shall have burrs removed. Threaded fasteners shall not be broken, cracked or stripped and shall be securely torqued. Rivets shall fill the hole completely and the heads shall be in full contact with the surface of the member and concentric with the hole.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 In-process examination. Examination shall be made of the following processes to establish conformance with specified requirements. Whenever nonconformance is noted, correction shall be made to the affected items and process:

(a) Circuit elements coated with varnish as required in 3.5.12, when applicable.

(b) Outer shell stiffened and reinforced, when applicable (see 3.5.1).

(c) Oven chamber lining(s) reinforced and without open joints (see 3.5.2).

4.2.3 End item inspection.

* 4.2.3.1 End item examination. The end item shall be examined in accordance with 4.2.3.1.1 and 4.2.3.1.2. The lot shall consist of all the ovens of the same type, style, and model offered for inspection at one time. The sample unit shall be one oven.

* 4.2.3.1.1 Visual examination. The end item shall be examined for the defects listed in table II. The inspection level shall be II and the acceptable quality level (AQL) shall be 2.5 major and 6.5 total (major and minor combined) defects per hundred units.

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*TABLE II. Visual defects

Examine	Defect	Classification Major	Minor
Finish	Not as specified.		X
	Enameled surfaces blistered, peeled, chipped, or area of no film.		X
	Evidence of discoloration, or stains.	X	
Construction and workmanship	Component missing.	X	
	Any burr, sliver, sharp edge, rough die, tool, gouge, trim or grind marks.	X	
	Any component fractured, buckled, dented, punctured, or malformed.	X	
Body	Not constructed as specified.	X	
	Not fabricated of material specified. (Where visual determination can be made.)	X	
Oven top	Not as specified.	X	
	Not provided with stainless steel top.	X	
Oven compartment	Evidence of open joints in linings.	X	
Doors	Not full width of the oven interior.	X	
	Door windows (when required) do not permit viewing of the center three shelves.	X	
	Door handle missing.	X	
	Lower door section of door not horizontally level with the oven deck when fully open.	X	
	Doors when fully open obstruct removal of shelf or shelf supports.	X	
Shelf and shelf support (as applicable)	Not removable.	X	
	Not furnished with a minimum of five shelves.	X	
	Shelf jams or buckles when inserted or withdrawn in shelf support.		X
	Shelf support not easily removable.		X
Vent	Vent not as specified (when provided).	X	

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TABLE II. Visual Defects (cont'd)

Examine	Defect	Classification Major	Minor
Gas burners	Not removable from front of oven.	X	
	Not protected or supported.	X	
	Oven blower not controlled by door switch (i.e.) does not shut off when door is open or turn on when door is closed.	X	
	Not located to prevent heat loss.	X	
	Not accessible for cleaning or servicing.	X	
	Gas burner valve orifice not fixed type.	X	
	Orifices not color coded.	X	
Lights	Not accessible for replacing.	X	
	Exterior light control missing.	X	
Interior lighting	Not recessed or protected in compartment.	X	
	Not located to provide maximum illumination.	X	
Indicator light	Does not indicate when component is energized.	X	
Thermostat	Improper range.	X	
	Not protected from direct heat.	X	
	Not protected from grease and moisture.	X	
Timer	Not provided with an alarm signal.	X	
Open or enclosed stand or legs (as applicable)	Open or enclosed stand extends beyond oven sides and front.		X
	Open or enclosed stand not as specified.	X	
	Open or enclosed stand not reinforced.	X	
	Legs not reinforced (when secured directly to oven bottom).	X	
	Leveling devices missing.	X	
Flexible gas line	Not type specified.	X	

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TABLE II. Visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Connector	Not provided.	X	
Welding (when applicable)	Not free of oxide, scale, paint, or grease.	X	
	Not continuous, sound, smooth, or free from porosity, cracks, incomplete fusion or deformation of material.	X	
	Flux (when used) not removed.		X
Thread fasteners and rivets	Missing, broken, stripped, or fractured.	X	
	Loose.		X
	Rivets not fully peened or upset.		X
	Rivets head(s) not in full contact with riveted members.		X
Marking (instructions)	Missing, incomplete, incorrect, or not legible.	X	
	Not securely attached to oven.	X	
Marking (identification)	Missing, incomplete, or not legible.		X
	Not securely attached to oven.		X

4.2.3.1.2 Dimensional examination. Inspection shall be made of ovens to determine compliance with dimensions specified. Any nonconformance with the specified requirements shall constitute a defect. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

* 4.2.3.2 End item testing. One end item of the lot shall be tested as specified in 4.4.3, 4.4.4, 4.4.5, and when applicable 4.4.6. Failure to conform with the tests shall be cause for rejection of the lot. Each production unit shall be tested as specified in 4.4.2. Failure to pass the test shall be cause for rejection of the unit. Units shall be tested as specified in 4.4.1 using an inspection level of S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units. The sample unit shall be one oven.

4.2.4 Packaging inspection. An examination shall be made to determine that the preservation, packaging, packing, and marking required by MIL-B-2316 are complied with.

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4.3 Certification examination. Certificates of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the specification requirements shall be examined and validated.

4.4 Methods of inspection.

* 4.4.1 Shelf structural strength. The shelf shall be held in its normal position in the oven and subjected to a uniformly distributed load of 20 pounds per square foot. The shelf shall be examined for deflection and dislodgement. After 2 minutes, the load shall be removed and the shelf examined for permanent deflection. Inability of the shelf to meet the load requirements specified in 3.5.4 shall constitute failure of this test.

* 4.4.2 Operational test. Each oven compartment shall be connected to the specified gas supply and all components and connections tested with liquid soap to determine compliance with the gas leakage requirements specified in 3.5.7.4. Burners shall be tested by ignition to determine that they function properly and controls and alarm signal shall be tested by turning off and on to determine compliance with the operational requirements of 3.5.7.4. Any leakage or any nonconformance with the operational requirements specified shall constitute failure of this test.

4.4.3 Oven thermostat accuracy. Each oven compartment shall be turned on, each thermostat shall be successively set at 250°F, 300°F, and 400°F, and the compartment temperature shall be measured to determine compliance with temperature requirements specified in 3.5.8. Inability to meet the thermostat accuracy requirements specified shall constitute failure of this test.

* 4.4.4 Preheat time with blower energized. Prior to testing the oven, the ambient temperature shall be 70°F plus or minus 5°F with the door closed. The thermostat, for one compartment shall be set to at least 400°F and the convection oven blower energized. The compartment temperature shall be read at 1 minute intervals. The time required to reach a temperature of 400°F shall be recorded to determine compliance with 3.5.8.1. When applicable, the test shall be repeated for the second compartment of type II ovens to determine compliance. Inability to meet the oven temperature requirements specified shall constitute failure of this test.

4.4.5 Circulation and heat distribution test. These tests shall be performed in conjunction with the preheat time tests specified in 4.4.4. Thermocouples shall be located on the top, middle, and bottom shelves in the following manner: Lines shall be drawn on each of the specified shelves, 2 inches in and parallel to the front, back, and each side; mid-point to mid-point of each side; and mid-point of the back to mid-point of the front. A thermocouple shall be located at each line intersection for a total of nine per shelf. All temperatures shall be read on a recording potentiometer. Use of deflection type instruments shall not be considered satisfactory. Mean temperature readings on each shelf, at one thermostat setting, shall not vary more than twenty-five degrees of each other. This test shall be conducted to determine compliance with the requirements specified in 3.5.7.1 and the circulation and heat distribution requirements specified in 3.5.6.4. Any nonconformance with the circulation and heat distribution requirements specified shall constitute failure of this test.

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* 4.4.6 Electromagnetic compatibility test. When electromagnetic compatibility is required (see 3.5.11), the oven shall be tested by the contractor in accordance with test Methods CE04, CE05 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the contractor or an independent testing agency. The contractor shall furnish the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan and the Electromagnetic Compatibility Test Report and the requirements meet MIL-STD-461. Inability of the oven to meet the requirements specified shall constitute failure of this test.

5. PREPARATION FOR DELIVERY

5.1 Level A, B, and C. Preservation, packaging, packing and marking for each level of protection shall be in accordance with the applicable requirements of MIL-B-2316 (see 6.2).

6. NOTES

6.1 Intended use. The baking and roasting gas convection ovens are intended for use in kitchens and bakeries where combined use and time savings are required.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, style, and model of oven required (see 1.2).
(c) When doors without windows are required (see 3.5.3).
(d) Electrical characteristics when other than specified (see 3.5.7).
(e) When electromagnetic compatibility is required (see 3.5.11).
(f) When fungus and moisture treatment of electrical components is required (see 3.5.12).
(g) Selection of applicable level of preservation and packaging and packing (see 5.1).
(h) When aluminized-coated steel or porcelain enamel on carbon steel for oven compartment lining is required (see 3.5.2).
(i) When aluminized-coated steel or porcelain enamel on carbon steel for vent or air baffle is required (see 3.5.2.1).
(j) When aluminized-coated steel or porcelain enamel on carbon steel for the door interior is required (see 3.5.3).

6.3 Contract data requirements. Any data items to be delivered under any contract for items covered by this specification should be specifically called for in the contract in accordance with the applicable regulations of the procuring activity.

* 6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

* 6.5 Deleted items. The style 2 and model A ovens have been deleted as they are no longer required.

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6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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